DBDA - Practical Machine learning

```
1. What is true about Machine Learning?
```

Answers

1. Machine Learning (ML) is that field of computer science.
2. ML is a type of artificial intelligence that extract patterns out of raw data by using an algorithm or method.
3. The main focus of ML is to allow computer systems learn from experience without being explicitly programmed or human intervention.
4. **All of the above.**

```
2. RNN stands for?
```

Answers

1. Receives neural networks
2. Report neural networks
3. Recording neural networks
4. **Recurrent neural network**

```
3. What is the use of data cleaning?
```

Answers

1. to remove the noisy data
2. correct the inconsistencies in data
3. transformations to correct the wrong data.
4. **All of the above**

```
4. How would you import a decision tree classifier in sklearn?
```

Answers

1. from sklearn.decision_tree import DecisionTreeClassifier
2. from sklearn.ensemble import DecisionTreeClassifier
3. **from sklearn.tree import DecisionTreeClassifier**
4. None of these

```
5. What is called coefficient of determination?
```

Answers

1. **R^2**
2. MAE
3. MSE
4. Accuracy

```
6. A Measure of goodness of fit for the estimated regression is the
```

Answers

1. Multiple coefficient of determination
2. mean square due to error
3. **mean Square due to regression**
4. none of the above

```
7. If I am using all features of my dataset and I achieve 100% accuracy on my
training set, but 70% on validation set, what should I look out for?
```

Answers

1. Underfitting
2. Nothing, the model is perfect
3. **Overfitting**
4. None of these

```
8. In which of the following scenario a gain ratio is preferred over Information
Gain?
```

Answers

1. **When a categorical variable has very large number of category**
2. When a categorical variable has very small number of category
3. Number of categories is the not the reason
4. None of these

```
9. What techniques can be used to improve the efficiency of apriori algorithm?
```

Answers

1. Hash-based techniques
2. Transaction Reduction
3. Partitioning
4. **All of the above**

```
10. SVM can be used to solve _______ problems.
```

Answers

1. Classification
2. Regression
3. **Both Classification and Regression**
4. Clustering

```
11. When using R, which of the following package is used for SVM?
```

Answers

1. b1072
2. **e1071**
3. c1071
4. d2012

```
12. Which of the following function is used for k-means clustering ?
```

Answers

1. **k-means**
2. k-mean
3. heatmap
4. None of the Mentioned

```
13. PCA is used for____.
```

Answers

1. Dimensionality Enhancement
2. **Dimensionality Reduction**
3. Both
4. None of above

```
14. Predictive Errors are due to
```

Answers

1. Bias
2. Variance
3. **Both of above**
4. None of above

```
15. white noise process will have
(i) A zero mean
(ii) A constant variance
(iii) Autocovariances that are constant
(iv) Autocovariances that are zero except at lag zero
```

Answers

1. **(ii) and (iv) only**
2. (i) and (iii) only
3. (i), (ii), and (iii) only
4. (i), (ii), (iii), and (iv)

```
16. Supervised learning can be split into subcategories
```

Answers

1. regression.
2. Classification
3. **Classification and regression both**
4. None of above.

```
17. One of the main challenge/s of NLP Is _____________
```

Answers

1. **Handling Ambiguity of Sentences**
2. Handling Tokenization
3. Handling POS-Tagging
4. All of the mentioned

```
18. Who is the “father” of artificial intelligence?
```

Answers

1. Fisher Ada
2. **John McCarthy**
3. Allen Newell
4. Alan Turning

```
19. Which of the following algorithm doesn’t uses learning Rate as of one of its
hyperparameter?
1.Gradient Boosting
2.Extra Trees
3.AdaBoost
4.Random Forest
```

Answers

1. 1 and 3
2. 1 and 4
3. 2 and 3
4. **2 and 4**

```
20. In random forest or gradient boosting algorithms, features can be of any
type. For example, it can be a continuous feature or a categorical feature.
Which of the following option is true when you consider these types of features?
```

Answers

1. Only Random forest algorithm handles real valued attributes by discretizing them
2. Only Gradient boosting algorithm handles real valued attributes by discretizing them
3. **Both algorithms can handle real valued attributes by discretizing them**
4. None of these

```
21. How the decision tree reaches its decision?
```

Answers

1. Single test
2. Two test
3. **Sequence of test**
4. No test

```
22. What does Apriori algorithm do?
```

Answers

1. **It mines all frequent patterns through pruning rules with lesser support**
2. It mines all frequent patterns through pruning rules with higher support
3. Both a and b
4. None of the above

```
23. When do you consider an association rule interesting?
```

Answers

1. If it only satisfies min_support
2. If it only satisfies min_confidence
3. **If it satisfies both min_support and min_confidence**
4. There are other measures to check so

```
24. K means and K-medioids are example of which type of clustering method?
```

Answers

1. Hierarchical
2. **partition**
3. probabilistic
4. None of the above.

```
25. Adding more basis functions in linear model
```

Answers

1. **Decreases model bias**
2. Decreases estimation bias
3. Decreases Variance
4. Doesn't affect bias and Variance

```
26. High entropy means that the partitions in classification are__
```

Answers

1. pure
2. **not pure**
3. useful
4. useless

```
27. Which of the following is NOT supervised learning?
```

Answers

1. **PCA**
2. Decision Tree
3. Linear Regression
4. Naive Bayesian

```
28. Which of the following statements about Naive Bayes is incorrect?
```

Answers

1. Attributes are equally important.
2. **Attributes are statistically dependent of one another given the class value.**
3. Attributes are statistically independent of one another given the class value.
4. Attributes can be nominal or numeric

```
29. Which of the following machine learning algorithm can be used for imputing
missing values of both categorical and continuous variables?
```

Answers

1. **K-NN**
2. Linear Regression
3. Logistic Regression
4. None of above

```
30. Which of the following distance measure do we use in case of categorical
variables in k-NN?
Hamming Distance
Euclidean Distance
Manhattan Distance
```

Answers

1. **1**
2. 2
3. 3
4. 2 and 3

```
31. Which of the following will be Euclidean Distance between the two data point
A(1,3) and B(2,3)?
```

Answers

1. **1**
2. 2
3. 4
4. 8

```
32. Father of Machine Learning (ML)
```

Answers

1. Geoffrey Chaucer
2. Geoffrey Hill
3. **Geoffrey Everest Hinton**
4. None of the above

```
33. What kind of learning algorithm for “Facial identities or facial
expressions”?
```

Answers

1. Prediction
2. **Recognition Patterns**
3. Generating Patterns
4. Recognizing Anomalies Answer

```
34. Targetted marketing, Recommended Systems, and Customer Segmentation are
applications in which of the following?
```

Answers

1. Supervised Learning: Classification
2. **Unsupervised Learning: Clustering**
3. Unsupervised Learning: Regression
4. Reinforcement Learning

```
35. Which of the following is a disadvantage of decision trees?
```

Answers

1. **Decision trees are prone to be overfit**
2. Decision trees are robust to outliers
3. Factor analysis
4. None of the above

```
36. Probability provides a way of summarizing the _____  that comes from our
laziness
```

Answers

1. Belief
2. **Uncertaintity**
3. Joint probability distributions
4. Randomness

```
37. The benefit of Naïve Bayes
```

Answers

1. Naïve Bayes is one of the fast and easy ML algorithms to predict a class
2. It is the most popular choice for text classification problems.
3. It can be used for Binary as well as Multi-class
4. **All of the above**

```
38. What is Gini Index?
```

Answers

1. **It is calculated by subtracting the sum of squared probabilities of each class from one**
2. It is used to measure the impurity or randomness of a dataset.
3. To find the best feature which serves as a root node

4. None of the options

```
39. What is Entropy?
```

Answers

1. It is calculated by subtracting the sum of squared probabilities of each class from one
2. **It is used to measure the impurity or randomness of a dataset.**
3. To find the best feature which serves as a root node
4. To find the best feature which serves as a root node

```
40. NLTK stands for ____
```

Answers

1. **Natural Language Toolkit**
2. Natural Language Transciption kit
3. Natural Learning Toolkit
4. Natural Learning Transciption kit